

TEST REPORT: TR-2026-894-ELEC

Product Under Test: Industrial Controller X200 (HW Rev 2.1)

Manufacturer: Atlas Motion Systems | Test Date: August 12-16, 2026

Accreditation: ISO/IEC 17025 Certified Electrical & Safety Test Lab

1. Executive Summary & Verdict Table

This compliance test report details the comprehensive verification of the Industrial Controller X200 against electrical safety, power boundaries, dielectric insulation, and timing criteria specified in SRS-2026-X200-EU and harmonized standard IEC 61010-1:2010.

Clause / Test Parameter	Required Spec	Measured Value	Verdict
Input Voltage Operating Range	18.0 V to 30.0 V DC	Tested at 18.0V, 24.0V, 30.0V	PASS
Max Power Consumption (Full Load)	· 45.0 W	38.6 W measured at 24V	PASS
Cold-Start Inrush Current	· 8.0 A, duration · 10 ms	6.2 A peak for 4.8 ms	PASS
Dielectric Withstand (Chassis/IO)	2.5 kV AC RMS for 60s	2.50 kV applied, leakage 0.825mA	PASS
STO Emergency Stop Latency	· 25.0 ms	18.4 ms measured	PASS
Thermal Operating Limit Test	-20.0 °C to +55.0 °C	Pass at -20°C, Pass at +55°C	PASS

2. Electrical Power and Voltage Tolerance Verification (REQ-ELE-001)

Test Procedure: The unit under test (EUT) was powered via a programmable DC power supply (Chroma 62050P). Input voltage was swept from 17.5 V DC to 31.0 V DC in increments of 0.5 V under continuous I/O cycling. The controller booted and operated without communication dropouts or memory resets from 18.0 V to 30.0 V DC. Under 24.0 V DC nominal supply, the measured steady-state power draw was 38.6 W with all 32 output channels active. Transient overvoltage pulse testing: A 36.0 V DC pulse of 500 ms duration was injected at 1-minute intervals. The internal TVS clamping diodes maintained internal rail stability with zero system resets recorded.

3. Dielectric Insulation & High-Pot Testing (REQ-ELE-003)

Test Standard: IEC 61010-1 Clause 6.8 (Dielectric Strength Test).

Test Equipment: QuadTech Guardian 6000 HiPot Tester (Cal due: 2027-03-15).

Test Points:

- Terminal Block A (24V Power In) to Protective Earth (Chassis Ground)
- Digital I/O Bus Terminals to Isolated MCU Logic Ground Plane

Test Voltage: 2,500 V AC RMS (50 Hz), ramp-up time 5.0 s, hold duration 60.0 seconds.

Results: Zero breakdown observed. Measured leakage current: 0.32 mA (allowable limit < 1.0 mA). Verdict: PASS.

4. Functional Safety STO Timing & E-Stop Latency (REQ-SAF-001)

Test Setup: Dual-channel safety relay input STO_A and STO_B were connected to a high-speed logic analyzer (Keysight Infiniium S-Series, 500 MSa/s). The motor drive PWM output gate signal was monitored across 50 simulated emergency stop events under full load.

Timing Results Summary:

- Mean STO Disengage Latency: 18.42 ms
- Maximum Recorded Latency: 21.15 ms (well below 25.0 ms threshold limit)
- Minimum Recorded Latency: 16.80 ms
- Channel Discrepancy Detection: 2.1 ms (switches to Safe State within 5 ms on single-channel failure)

Verdict: Fully compliant with ISO 13849-1 Category 3 PL d and IEC 61508 SIL 2 requirements.